Do Frozen SMILES Foundation Embeddings Beat Fingerprints?
A Split- and Low-Data Benchmark on ADMET and CNS Bioactivity Tasks

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Abstract

Frozen molecular embeddings are a common low-cost way to use pretrained SMILES models, but the practical question is whether these embeddings beat strong fingerprint baselines under realistic split and label-budget constraints. We benchmark cached frozen ChemBERTa and MoLFormer embeddings against Morgan fingerprints and RDKit descriptors on ADMET and CNS bioactivity classification tasks. To avoid conflating representation quality with downstream classifier choice, each representation family is paired with logistic regression, random forest, and XGBoost classifiers where available. The revised local-v3 benchmark contains 2580 completed fits across 8 curated binary tasks, 12 model configurations, random splits, scaffold splits, and low-data training budgets of 50, 100, 250, and 500 molecules. Frozen-vs-classical family comparisons are available for 43 dataset–regime rows; in this retrospective oracle comparison, the best classical baseline wins 36 rows and the best frozen embedding wins 7 rows. Across 215 paired seed-level comparisons, the mean AUROC difference for best frozen embedding minus best classical baseline is -0.026 with a 95% bootstrap confidence interval of [-0.032, -0.021]. A clustered bootstrap over dataset–regime rows gives a similar mean difference of -0.025 [-0.032, -0.018]. The average gap is modest but consistently non-positive across most regimes. These results support a narrow, practical recommendation: frozen SMILES foundation embeddings should not be adopted by default without first benchmarking against strong Morgan/RDKit baselines under the intended split protocol and label budget.

1 Introduction

Molecular property prediction has benefited from public benchmarks such as MoleculeNet [Wu et al., 2018] and Therapeutics Data Commons [Huang et al., 2021]. In parallel, pretrained molecular models have created enthusiasm around transferable representations. ChemBERTa [Chithrananda et al., 2020] and MoLFormer [Ross et al., 2022] learn from large SMILES corpora, while graph and 3D systems such as GROVER [Rong et al., 2020] and Uni-Mol [Zhou et al., 2023] represent different model families.

This study deliberately asks a narrower question than whether all molecular foundation models are generally superior. We evaluate a practical deployment pattern: extract frozen ChemBERTa or MoLFormer embeddings, then train a cheap classifier. The relevant baseline is not weak. Extended-connectivity fingerprints [Rogers and Hahn, 2010], RDKit descriptors [RDKit contributors, 2026], random forests, and XGBoost [Chen and Guestrin, 2016] remain strong in QSAR practice [Muratov et al., 2020]. A frozen embedding is useful in this setting only if it improves over these baselines under the split protocol and label budget that a practitioner will actually use.

Figure 1: Benchmark workflow. Public molecular datasets are curated into binary tasks, represented either by classical Morgan/RDKit features or frozen ChemBERTa/MoLFormer embeddings, paired with the same downstream classifier families, and evaluated under random, scaffold, and low-data split protocols.

2 Related Work

Large molecular benchmarks usually emphasize broad task coverage and aggregate leaderboards [Wu et al., 2018, Huang et al., 2021]. Small-data molecular learning has also been treated as a distinct challenge because high-capacity models can overfit scarce assay labels [Dou et al., 2023]. Transfer learning methods such as MolPMoFiT illustrate how pretrained representations can help after adaptation [Li and Fourches, 2020], and recent uncertainty work highlights the importance of calibration in ADMET-style deployment [Li et al., 2024].

The closest prior work is the benchmark of pretrained molecular embedding models by Praski et al. [2025], which evaluates 25 models on 25 datasets and uses hierarchical Bayesian statistical testing. That work finds that most neural pretrained embeddings do not clearly improve over ECFP baselines, with CLAMP standing out as a strong exception. Our contribution is therefore not to claim that this phenomenon is new. Instead, we provide a lightweight, reproducible audit focused on frozen ChemBERTa and MoLFormer embeddings, explicit 50/100/250/500-label regimes, ADMET plus CNS ChEMBL tasks [Zdrazil et al., 2023], classifier-controlled comparisons, calibration summaries, and downstream runtime. Thus, the intended contribution is replication and practical audit rather than model-scale comprehensiveness. Adjacent 2026 work on tabular foundation models for molecular properties also emphasizes the interaction between representation choice, frozen embeddings, descriptors, and low- to medium-data prediction [Ben Hicham et al., 2026]. Scaffold and out-of-distribution evaluation remain important because chemical extrapolation is often weaker than in-distribution performance [Antoniuk et al., 2025].

3 Benchmark Design

3.1 Dataset Curation

Table 1 summarizes the curated local-v3 tasks. SMILES strings are parsed and canonicalized with RDKit before duplicate handling. Canonical-SMILES groups with conflicting binary labels are removed, while non-conflicting duplicate groups are collapsed to the first retained row. ChEMBL CNS rows are first converted to target-specific binary tasks using a pChEMBL ≥ 6.0 activity cutoff and then curated by target. We do not perform salt stripping, neutralization, stereochemistry normalization beyond RDKit canonical SMILES, or tautomer canonicalization; this is a deliberate limitation. All splits are created after this curation step.

Table 1: Dataset curation summary. Duplicate and conflict counts are computed after RDKit canonical-SMILES normalization.

Dataset	Raw rows	Valid SMILES	Unique canonical	Duplicate rows	Conflict groups	Final n	Positives	Positive rate
BBB Martins	2039	2039	1975	64	10	1965	1500	76.34%
B3DB	6167	6167	6167	0	0	6167	3989	64.68%
HIA Hou	578	578	578	0	0	578	500	86.51%
Bioavailability Ma	640	640	640	0	0	640	492	76.88%
PAMPA NCATS	2035	2035	2034	1	0	2034	1739	85.50%
ChEMBL217 D2	391	391	332	59	1	331	184	55.59%
ChEMBL228 SERT	365	365	263	102	18	245	148	60.41%
ChEMBL233 MOR	399	399	385	14	1	384	204	53.12%

3.2 Representations, Classifiers, and Splits

Classical representations are Morgan fingerprints (radius 2, 1024 bits) and a compact RDKit descriptor block. The descriptor block contains molecular weight, MolLogP, TPSA, heavy-atom count, hydrogen-bond donor and acceptor counts, rotatable-bond count, ring count, aromatic and aliphatic ring counts, fraction sp^3 carbon, BertzCT, BalabanJ, and LabuteASA. Frozen SMILES embeddings are cached 768-dimensional ChemBERTa and MoLFormer vectors generated for all eight curated tasks. Each representation family is paired with logistic regression (LR), random forest (RF), and XGBoost (XGB). This produces Morgan LR/RF/XGB, RDKit LR/RF/XGB, ChemBERTa LR/RF/XGB, and MoLFormer LR/RF/XGB. LR models use standardized features; RF and XGB use the raw representation values.

Table 2: Frozen embedding extraction details. Both encoders are used as frozen feature extractors; no task labels are used during embedding generation.

Encoder	Checkpoint	Pooling	Max length	Batch size	Device / notes
ChemBERTa	`seyonec/ChemBERTa-zinc-base-v1`	Attention-mask mean over last hidden state	256	128	CUDA; checkpoint tokenizer; truncation enabled
MoLFormer	`ibm-research/MoLFormer-XL-both-10pct`	Attention-mask mean over last hidden state	202	64	CUDA; deterministic random-feature attention, seed 0

Table 3: Embedding regeneration cost and tokenizer truncation diagnostics from the local-v3 cache metadata. Times are GPU wall-clock feature-extraction times on the local CUDA environment and are reported separately from downstream classifier runtime.

Encoder	Molecules	Total time (s)	Molecules/s	Device	Disk (MB)	Max tokens	Truncated molecules
ChemBERTa	12614	21.374	590.145	cuda	36.956	306	2 (0.02%)
MoLFormer	12614	25.322	498.135	cuda	36.956	247	9 (0.07%)

Table 4: Downstream classifier hyperparameters. Hyperparameters are fixed across datasets and representations rather than tuned per task.

Classifier	Fixed settings
LR	`penalty=l2`, `C=1.0`, `solver=lbfgs`, `max_iter=2000`, `class_weight=balanced`; standardized inputs
RF	`n_estimators=128`, `max_features=sqrt`, `min_samples_leaf=1`, no max-depth cap, `class_weight=balanced_subsample`
XGB	`n_estimators=160`, `max_depth=4`, `learning_rate=0.05`, `subsample=0.85`, `colsample_bytree=0.70`, `tree_method=hist`, `eval_metric=logloss`; no explicit `scale_pos_weight`

Random splits use an 80/20 stratified split. Scaffold splits use RDKit Bemis–Murcko scaffolds [Bemis and Murcko, 1996]; molecules with an empty Murcko scaffold fall back to their canonical molecule string as the scaffold key. For each seed, 256 scaffold-group shuffles are scored, and the class-diverse split closest to the target 20% test size and global class balance is selected. If no class-diverse scaffold split exists, the code falls back to a stratified random split and records a warning. Low-data regimes are stratified subsamples from the random-split training pool with at least five molecules per class; unavailable budgets are skipped. Seeds 0–4 control splitting, low-data sampling, and stochastic classifier states.

Table 5: Scaffold split diagnostics across five seeds. Fallback count is the number of seeds where the scaffold splitter could not produce a class-diverse test fold and reverted to stratified random splitting.

Dataset	Scaffolds	Largest frac.	Mean test n	Test positive rate	Test scaffolds	Fallback count
B3DB	2247	0.059	1233.400	0.647	419.400	0
BBB Martins	1114	0.065	393.000	0.763	225.000	0
Bioavailability Ma	456	0.077	129.000	0.767	91.200	0
ChEMBL217 D2	164	0.079	66.000	0.561	34.200	0
ChEMBL228 SERT	101	0.127	49.000	0.612	20.800	0
ChEMBL233 MOR	141	0.057	77.200	0.531	32.000	0
HIA Hou	403	0.111	116.000	0.862	80.200	0
PAMPA NCATS	1388	0.026	407.000	0.855	278.600	0

The primary discrimination metric is AUROC. We also report AUPRC, MCC, F1, balanced accuracy, Brier score, log loss, 10-bin expected calibration error (ECE), and downstream CPU fit-plus-predict time. Runtime excludes embedding generation because the benchmark uses cached embeddings; Table 3 reports embedding regeneration separately. XGB is intentionally left without explicit class weighting, whereas LR and RF use class-weighted objectives; this choice avoids adding another representation-specific imbalance policy but should be considered when interpreting highly imbalanced tasks.

4 Results

4.1 Best Fingerprint/Descriptor Baselines Still Win Most Rows

Table 6 counts dataset–regime rows where both frozen embeddings and classical baselines are available. In local-v3, ChemBERTa and MoLFormer embeddings are cached for all eight curated tasks, so all 43 available dataset–regime rows are family-comparable; missing heatmap cells correspond to unavailable low-data budgets for small tasks. The comparison is retrospective and oracle-style within each family because it selects the best model after observing test performance. Under that convention, the best fingerprint/descriptor baseline wins 36 of 43 comparable rows, while the best frozen embedding wins 7 rows. The seven frozen wins are generally small and task-specific, rather than evidence of a broad frozen-embedding advantage.

Table 6: Winner counts among family-comparable dataset–regime rows. Difference is best frozen embedding AUROC minus best classical AUROC, averaged across rows in the regime.

Regime	Rows	Classical wins	Frozen wins	Mean difference
random	8	7	1	-0.034
scaffold	8	5	3	-0.011
low-data 50	8	8	0	-0.022
low-data 100	8	7	1	-0.031
low-data 250	7	5	2	-0.025
low-data 500	4	4	0	-0.033

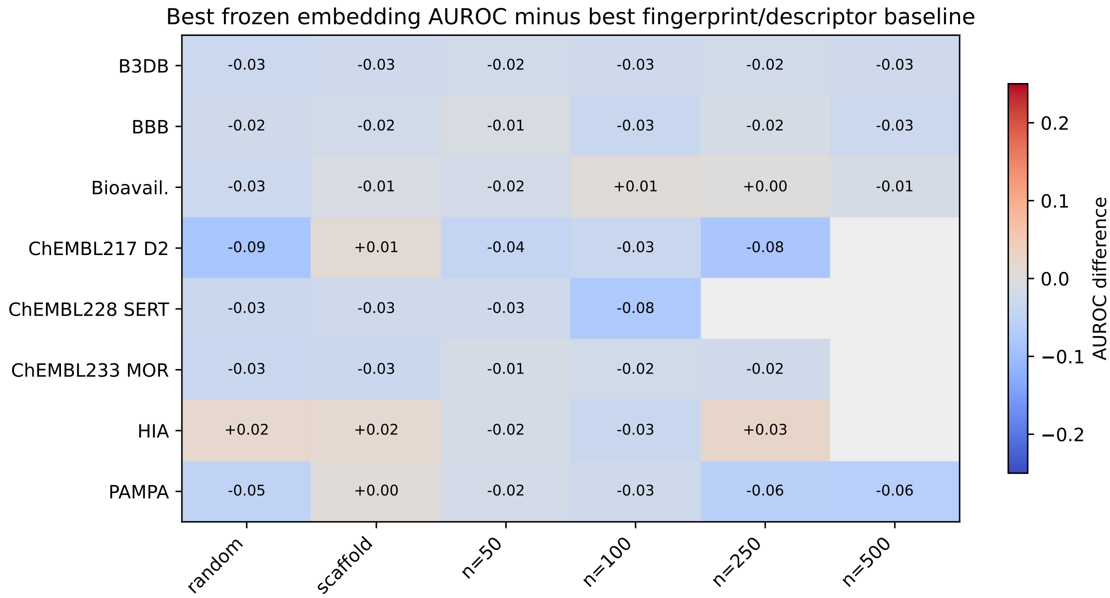

Figure 2: Task-level AUROC difference between the best frozen SMILES embedding model and the best Morgan/RDKit baseline across the 43 family-comparable rows. Positive values indicate a frozen-embedding advantage. Gray cells indicate unavailable low-data budgets rather than classical-only comparisons. Dataset labels are shortened for readability.

4.2 Paired Statistical Diagnostics

Winner counts are useful but incomplete because rows share datasets and seeds. Table 7 therefore reports paired seed-level comparisons. For each dataset, split, and seed, we take the best frozen embedding model and subtract the best classical baseline. Negative values favor Morgan/RDKit baselines. Across all 215 paired seed-level units, the mean difference is -0.026 AUROC with a 95% bootstrap confidence interval of [-0.032, -0.021]. Because seed-level units from the same dataset–regime row are not fully independent, Table 8 adds a clustered bootstrap over dataset–regime rows. Wilcoxon signed-rank tests are reported as diagnostics, not as a full hierarchical model.

Table 7: Paired AUROC diagnostics for best frozen embedding minus best classical baseline. Confidence intervals are nonparametric bootstrap intervals over paired seed-level units.

Regime	Paired units	Mean difference [95% CI]	Wilcoxon p
random	40	-0.033 [-0.044, -0.022]	$< 10^{-4}$
scaffold	40	-0.017 [-0.030, -0.004]	0.0090
low-data 50	40	-0.018 [-0.034, -0.003]	0.0111
low-data 100	40	-0.031 [-0.043, -0.019]	$< 10^{-4}$
low-data 250	35	-0.028 [-0.043, -0.014]	0.0006
low-data 500	20	-0.035 [-0.047, -0.025]	$< 10^{-4}$
all	215	-0.026 [-0.032, -0.021]	$< 10^{-4}$

Table 8: Clustered AUROC bootstrap diagnostics over dataset–regime rows for the retrospective oracle family comparison. Negative differences favor Morgan/RDKit baselines.

Regime	Rows	Mean difference [95% CI]	Classical win rows	Frozen win rows
random	8	-0.034 [-0.052, -0.014]	7	1
scaffold	8	-0.011 [-0.022, 0.002]	5	3
low-data 50	8	-0.022 [-0.029, -0.016]	8	0
low-data 100	8	-0.031 [-0.047, -0.017]	7	1
low-data 250	7	-0.025 [-0.050, -0.001]	5	2
low-data 500	4	-0.033 [-0.053, -0.018]	4	0
all	43	-0.025 [-0.032, -0.018]	36	7

4.3 Classifier-Controlled Comparisons

The original benchmark design could be criticized for comparing tree models on fingerprints against linear models on embeddings. Table 9 addresses this by comparing the best classical and best frozen representation under the same classifier family. The direction remains negative for LR, RF, and XGB. This does not prove that frozen embeddings are intrinsically weaker in all settings, but it weakens the simpler explanation that the headline result is only a tree-versus-linear artifact.

Table 9: Same-classifier paired AUROC comparisons across all family-comparable seed-level units. Negative differences favor Morgan/RDKit representations.

Classifier	Paired units	Mean difference [95% CI]	Classical wins	Frozen wins	Wilcoxon p
LR	215	-0.018 [-0.026, -0.010]	131	84	$< 10^{-4}$
RF	215	-0.031 [-0.037, -0.025]	177	38	$< 10^{-4}$
XGB	215	-0.019 [-0.026, -0.012]	146	69	$< 10^{-4}$

Because family-best selection is an oracle analysis, Table 10 also reports four pre-specified model-pair comparisons. Morgan RF remains stronger than ChemBERTa RF and ChemBERTa XGB on clustered mean difference, while Morgan XGB and ChemBERTa XGB are approximately tied. The MoLFormer RF comparison is clearly unfavorable to the frozen embedding in this local-v3 setting.

Table 10: Pre-specified model-pair comparisons. Differences are frozen model AUROC minus baseline model AUROC; clustered intervals are over dataset–regime rows.

Comparison	Rows	Clustered mean difference [95% CI]	Baseline seed wins	Frozen seed wins	Wilcoxon p
Morgan RF vs ChemBERTa RF	43	-0.018 [-0.027, -0.010]	156	59	$< 10^{-4}$
Morgan RF vs ChemBERTa XGB	43	-0.020 [-0.030, -0.010]	153	62	$< 10^{-4}$
Morgan XGB vs ChemBERTa XGB	43	0.001 [-0.011, 0.014]	111	104	0.6910
Morgan RF vs MoLFormer RF	43	-0.072 [-0.086, -0.058]	190	25	$< 10^{-4}$

4.4 Model-Level Performance, Calibration, and Runtime

Table 11 gives model-level means on family-comparable rows. Morgan RF has the highest mean AUROC, while ChemBERTa RF and ChemBERTa XGB are the strongest frozen-embedding variants. The calibration results should be read narrowly: before any post-hoc Platt, isotonic, or temperature calibration, classical baselines have lower family-level mean Brier score and ECE in this benchmark, although individual frozen configurations such as ChemBERTa RF can be competitive. This is not a claim that fingerprint models are intrinsically better calibrated under calibration-controlled evaluation.

Table 11: Model-level summary on rows where both representation families are available. Time is median downstream CPU fit-plus-predict seconds and excludes embedding generation.

Model	Family	Representation	Classifier	AUROC	Brier	ECE	Time (s)
Morgan RF	classical	morgan	RF	0.795 ± 0.110	0.137 ± 0.038	0.081 ± 0.041	0.208
RDKit RF	classical	rdkit	RF	0.788 ± 0.095	0.143 ± 0.044	0.083 ± 0.051	0.206
ChemBERTa RF	foundation	chemberta	RF	0.777 ± 0.109	0.145 ± 0.041	0.074 ± 0.038	0.210
ChemBERTa XGB	foundation	chemberta	XGB	0.775 ± 0.108	0.155 ± 0.052	0.115 ± 0.054	1.466
Morgan XGB	classical	morgan	XGB	0.775 ± 0.118	0.145 ± 0.046	0.096 ± 0.048	0.101
RDKit XGB	classical	rdkit	XGB	0.774 ± 0.101	0.154 ± 0.053	0.105 ± 0.060	0.042
Morgan LR	classical	morgan	LR	0.763 ± 0.106	0.175 ± 0.054	0.162 ± 0.055	0.021
ChemBERTa LR	foundation	chemberta	LR	0.753 ± 0.116	0.197 ± 0.072	0.187 ± 0.076	0.035
RDKit LR	classical	rdkit	LR	0.724 ± 0.114	0.196 ± 0.044	0.178 ± 0.067	0.009
MoLFormer RF	foundation	molformer	RF	0.723 ± 0.101	0.164 ± 0.046	0.086 ± 0.042	0.212
MoLFormer XGB	foundation	molformer	XGB	0.720 ± 0.098	0.175 ± 0.059	0.126 ± 0.063	1.437
MoLFormer LR	foundation	molformer	LR	0.715 ± 0.089	0.220 ± 0.050	0.212 ± 0.058	0.046

Table 12: Family-level calibration and downstream runtime summary on family-comparable rows. These are uncalibrated probabilities.

Family	AUROC	Brier	ECE	Median time (s)
classical	0.770 ± 0.110	0.158 ± 0.051	0.117 ± 0.066	0.066
foundation	0.744 ± 0.107	0.176 ± 0.060	0.133 ± 0.076	0.205

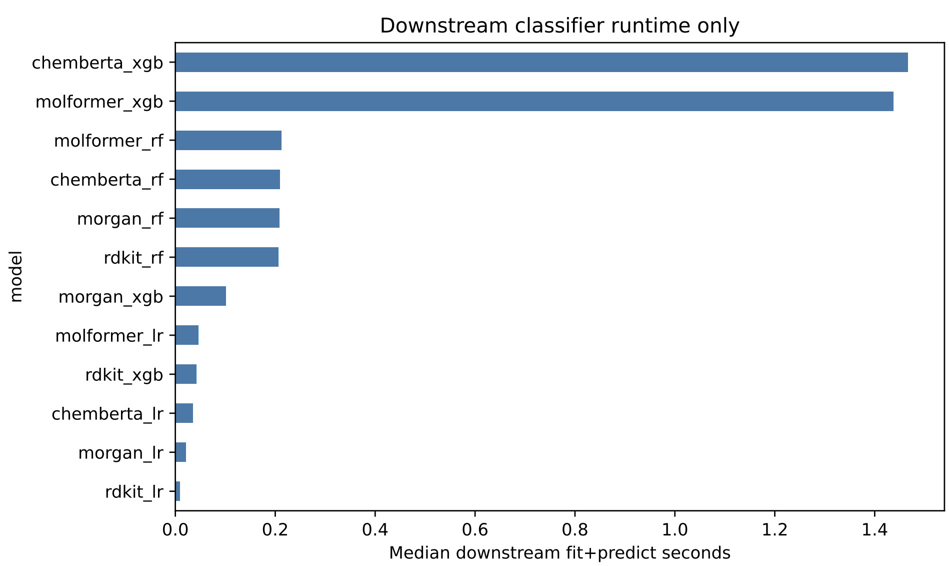

Figure 3: Median downstream classifier runtime by model. The plot excludes the GPU and storage cost of generating ChemBERTa or MoLFormer embeddings, so it should not be interpreted as end-to-end foundation-model deployment cost.

5 Failure Modes and Practical Recommendations

Three patterns are most relevant for practitioners. First, scaffold shift does not reliably rescue frozen SMILES embeddings; several scaffold rows are close or slightly positive, but the scaffold clustered interval overlaps zero and the overall direction remains classical-favoring. Second, low-data regimes remain task-specific. Frozen embeddings can be competitive, especially on selected HIA and bioavailability rows, but the best Morgan/RDKit baseline remains stronger on average at 50, 100, 250, and 500 labels. Third, classifier choice matters, yet the same-classifier analysis still favors the classical representation family on average.

The average gap is modest, so the conclusion is not that frozen embeddings are unusable. Rather, they did not provide a reliable default advantage over cheaper fingerprint baselines in this frozen-feature workflow.

The resulting recommendation is conservative: always run Morgan fingerprint and RDKit descriptor baselines before adopting frozen SMILES foundation embeddings. Frozen ChemBERTa is worth testing, especially with RF or XGB heads, but its benefit should be treated as a task-level empirical question. Fine-tuning large encoders may improve results, but it changes the resource profile and is not the low-cost frozen-feature workflow studied here.

6 Practical Benchmarking Checklist

For molecular benchmarks where frozen embeddings are used as plug-in features, we recommend the following minimum practice: include Morgan fingerprint RF/XGB baselines; report random and scaffold splits separately; evaluate low-data budgets when assay labels are scarce; compare representations under the same classifier family; report calibration before and after post-hoc calibration if probabilities drive decisions; and report embedding generation cost separately from downstream classifier runtime. This checklist is intentionally modest, but it prevents common over-interpretations of frozen embedding results.

7 Limitations

This benchmark is intentionally narrow. It evaluates frozen ChemBERTa and MoLFormer embeddings, not fine-tuned encoders, Uni-Mol, GROVER, GIN-pretrained models, CLAMP, CheMeleon, or tabular foundation model predictors. It covers binary ADMET and CNS bioactivity tasks, not regression, multitask toxicity panels, temporal assay shift, or property-based OOD splits. Runtime excludes embedding generation. Calibration is reported only before post-hoc calibration. Finally, paired Wilcoxon tests, seed-level bootstrap intervals, clustered bootstrap intervals, and leave-one-dataset-out sensitivity checks do not replace a full hierarchical Bayesian or Bradley–Terry analysis; they are included to provide transparent statistical diagnostics.

8 Code and Data Availability

Code and result artifacts are available at the public GitHub repository. A frozen reproducibility release corresponding to this arXiv version is archived on Zenodo with DOI 10.5281/zenodo.20142891 [Sriwicha, 2026]. The release includes the benchmark harness, generated result tables, figure assets, embedding-cache metadata, and instructions for regenerating ChemBERTa and MoLFormer embeddings from public input datasets. Public source datasets remain attributable to their original providers, including TDC task wrappers, B3DB, and ChEMBL. Raw datasets, API keys, model checkpoints, and cached embedding arrays are not redistributed in the GitHub repository; embedding caches can be regenerated from the documented public inputs.

9 Conclusion

In this local-v3 benchmark, cached frozen ChemBERTa and MoLFormer embeddings do not consistently outperform Morgan/RDKit baselines across curated ADMET and CNS bioactivity tasks, even when both representation families are paired with LR, RF, and XGB heads. The strongest classical baseline wins 36 of 43 family-comparable rows, and paired seed-level plus clustered diagnostics favor the classical baseline on average. Frozen SMILES foundation embeddings should therefore be treated as candidate representations, not default replacements for strong fingerprint baselines.

A Dataset Sensitivity

Table 13 summarizes the average oracle family difference per dataset and the leave-one-dataset-out overall mean. All leave-one-dataset-out means remain negative.

Table 13: Dataset-level sensitivity for best frozen embedding minus best classical AUROC.

Dataset	Dataset mean difference	Overall mean when excluded
B3DB	-0.026	-0.027
BBB Martins	-0.023	-0.027
Bioavailability Ma	-0.005	-0.030
ChEMBL217 D2	-0.051	-0.023
ChEMBL228 SERT	-0.033	-0.026
ChEMBL233 MOR	-0.028	-0.026
HIA Hou	-0.010	-0.029
PAMPA NCATS	-0.039	-0.024

B Full Dataset–Regime Results

Table 14: Full row-level oracle comparison. Difference is best frozen embedding AUROC minus best classical AUROC.

Dataset	Regime	Best classical	Classical AUROC	Best frozen	Frozen AU-ROC	Difference
B3DB	low-data 100	RDKit RF	0.811	ChemBERTa RF	0.785	-0.026
B3DB	low-data 250	RDKit RF	0.837	ChemBERTa RF	0.814	-0.023
B3DB	low-data 50	RDKit RF	0.791	ChemBERTa RF	0.768	-0.023
B3DB	low-data 500	Morgan RF	0.871	ChemBERTa XGB	0.846	-0.026
B3DB	random	Morgan RF	0.942	MoLFormer RF	0.911	-0.031
B3DB	scaffold	Morgan RF	0.892	ChemBERTa RF	0.865	-0.026
BBB Martins	low-data 100	RDKit RF	0.856	ChemBERTa LR	0.821	-0.034
BBB Martins	low-data 250	Morgan RF	0.860	ChemBERTa RF	0.845	-0.015
BBB Martins	low-data 50	RDKit RF	0.812	ChemBERTa LR	0.800	-0.011
BBB Martins	low-data 500	Morgan RF	0.900	ChemBERTa XGB	0.869	-0.031
BBB Martins	random	Morgan RF	0.923	ChemBERTa RF	0.899	-0.024
BBB Martins	scaffold	Morgan RF	0.900	ChemBERTa XGB	0.878	-0.022
Bioavailability Ma	low-data 100	RDKit RF	0.644	ChemBERTa XGB	0.650	0.006
Bioavailability Ma	low-data 250	Morgan RF	0.683	ChemBERTa XGB	0.684	0.000
Bioavailability Ma	low-data 50	RDKit RF	0.632	MoLFormer LR	0.614	-0.018
Bioavailability Ma	low-data 500	Morgan LR	0.717	ChemBERTa XGB	0.703	-0.014
Bioavailability Ma	random	Morgan RF	0.734	ChemBERTa RF	0.703	-0.031
Bioavailability Ma	scaffold	RDKit RF	0.715	ChemBERTa RF	0.702	-0.013
ChEMBL217 D2	low-data 100	Morgan RF	0.748	ChemBERTa RF	0.714	-0.034

Dataset	Regime	Best classical	Classical AUROC	Best frozen	Frozen AU-ROC	Difference
ChEMBL217 D2	low-data 250	Morgan RF	0.839	ChemBERTa RF	0.757	-0.082
ChEMBL217 D2	low-data 50	Morgan RF	0.741	ChemBERTa RF	0.698	-0.043
ChEMBL217 D2	random	Morgan RF	0.847	ChemBERTa RF	0.761	-0.086
ChEMBL217 D2	scaffold	Morgan RF	0.711	ChemBERTa RF	0.725	0.013
ChEMBL228 SERT	low-data 100	Morgan RF	0.861	ChemBERTa RF	0.784	-0.077
ChEMBL228 SERT	low-data 50	Morgan RF	0.831	ChemBERTa RF	0.805	-0.026
ChEMBL228 SERT	random	Morgan LR	0.907	ChemBERTa XGB	0.874	-0.032
ChEMBL228 SERT	scaffold	Morgan RF	0.852	ChemBERTa XGB	0.827	-0.025
ChEMBL233 MOR	low-data 100	Morgan RF	0.879	ChemBERTa XGB	0.856	-0.023
ChEMBL233 MOR	low-data 250	Morgan XGB	0.909	ChemBERTa XGB	0.886	-0.023
ChEMBL233 MOR	low-data 50	Morgan XGB	0.848	ChemBERTa RF	0.833	-0.015
ChEMBL233 MOR	random	Morgan XGB	0.916	ChemBERTa XGB	0.881	-0.034
ChEMBL233 MOR	scaffold	Morgan RF	0.897	ChemBERTa XGB	0.864	-0.033
HIA Hou	low-data 100	RDKit LR	0.877	ChemBERTa RF	0.843	-0.034
HIA Hou	low-data 250	RDKit RF	0.878	ChemBERTa LR	0.904	0.025
HIA Hou	low-data 50	RDKit RF	0.842	ChemBERTa RF	0.824	-0.017
HIA Hou	random	Morgan XGB	0.923	ChemBERTa XGB	0.942	0.019
HIA Hou	scaffold	Morgan LR	0.917	ChemBERTa LR	0.933	0.016
PAMPA NCATS	low-data 100	RDKit XGB	0.652	MoLFormer XGB	0.625	-0.027
PAMPA NCATS	low-data 250	RDKit RF	0.694	ChemBERTa RF	0.634	-0.060
PAMPA NCATS	low-data 50	RDKit RF	0.644	MoLFormer XGB	0.625	-0.018
PAMPA NCATS	low-data 500	RDKit RF	0.728	ChemBERTa XGB	0.666	-0.063
PAMPA NCATS	random	RDKit RF	0.756	ChemBERTa RF	0.705	-0.050
PAMPA NCATS	scaffold	RDKit XGB	0.724	ChemBERTa XGB	0.728	0.004

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